

E031

BSCNE-Azure Capstone Project

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word _____	The one fact that decides _____	Answer _____	[] sure [] unsure [] guess
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Items

1.

You create a /27 subnet in a VNet for web servers. The VNet is in the East US region and uses a custom DNS server. How many usable addresses does the subnet provide?

- A. 32
- B. 30
- C. 27
- D. 26

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2. [TRANSFER — reversed stem / different variable]

You need an Azure subnet for 50 VMs. What is the smallest prefix that works?

- A. /29
- B. /25
- C. /26
- D. /27

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

An NSG has priority 200: deny all inbound from 203.0.113.0/24, and priority 300: allow inbound TCP 443 from any source. The NSG was created last week and is attached to a subnet with four VMs. A client at 203.0.113.50 opens HTTPS to a VM. What happens?

- A. Denied by the priority 200 rule.
- B. Denied by the default DenyAllInbound rule.
- C. Allowed by the priority 300 rule.
- D. Allowed, because allow rules win over deny rules.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4. [TRANSFER — reversed stem / different variable]

An NSG denies SSH from the internet at priority 200. You must allow SSH from one admin IP range. What priority should the new allow rule have?

- A.** 65001, next to the default rules.
- B.** Exactly 200.
- C.** A number higher than 200, such as 250.
- D.** A number lower than 200, such as 150.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5.

A help-desk analyst must restart and resize VMs in the rg-web resource group only. A teammate proposes Owner at the subscription so the analyst “never gets blocked.” The analyst works the night shift. What assignment fits?

- A.** Reader at rg-web.
- B.** Virtual Machine Contributor at the rg-web resource group.
- C.** Owner at the subscription.
- D.** Contributor at the management group.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6. [TRANSFER — reversed stem / different variable]

A user has Contributor on resource group rg-app only. Can the user modify a storage account in rg-data in the same subscription?

- A.** Only if the storage account is in the same region.
- B.** No; the role applies only to rg-app and its resources.
- C.** Yes; Contributor covers the whole subscription.
- D.** Yes; resource groups in one subscription share permissions.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

A web app in Azure App Service reads from Azure Key Vault. The current code stores a client secret in a config file. The app is written in C# and deployed from a pipeline. What is the better approach?

- A. Rewrite the app in another language.
- B. Encrypt the config file but keep the secret.
- C. Enable a managed identity for the app and grant it access to the vault.
- D. Move the secret into the source code.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8.

A resource group holds a database, two VMs, and a storage account. Your Bicep template defines only the two VMs. You deploy it in complete mode. The template file is 90 lines long. What happens?

- A. The database and storage account are deleted.
- B. All resources, including the VMs, are deleted.
- C. Only the VMs are updated; the rest are untouched.
- D. The deployment fails because the template is too short.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9.

Company rules say all resources must be in approved US regions and carry a cost-center tag. There are 30 people with Contributor rights, and deployments run through two pipelines. What is the right tool to enforce the rules?

- A. Remove Contributor from all 30 people.
- B. Azure Policy with allowed-locations and required-tag rules.
- C. A custom RBAC role.
- D. Merge into one pipeline.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10.

VNet-A is peered with VNet-Hub, and VNet-Hub is peered with VNet-C. No gateway or network appliance routes traffic in the hub. The hub VNet is the oldest of the three. A VM in VNet-A cannot reach a VM in VNet-C. Why?

- A. Peering never allows VM-to-VM traffic.
- B. The NSG on VNet-C must be deleted.
- C. The hub is too old.
- D. Peering is non-transitive; A and C need a direct peering or hub routing.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11. [TRANSFER — reversed stem / different variable]

VNet-A and VNet-C must communicate, and the hub has no routing appliance. Which change makes it work?

- A. Create a direct peering between VNet-A and VNet-C.
- B. Delete the hub peerings.
- C. Move the VMs to the same region.
- D. Add a second peering between A and the hub.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12.

Someone deleted a load balancer yesterday. The team's Azure Monitor dashboard shows CPU and network graphs, and Defender for Cloud gives the subscription a secure score of 72%. Where do you find who deleted it?

- A. The VM's guest OS event log.
- B. Azure Monitor metrics.
- C. Defender for Cloud secure score.
- D. The Azure Activity log.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer C Concept 1

You create a /27 subnet in a VNet for web servers. ~~The VNet is in the East US region and uses a custom DNS server.~~ How many usable addresses does the subnet provide?

Target word: usable addresses in an Azure subnet **Decides:** 32 – 5 reserved.

A	Dropped element	Leaves out reserved addresses.
B	Adjacent category	Traditional minus-2 rule, not Azure.
C	Correct	$2^5 = 32$; Azure reserves 5.
D	Camouflage	Subtracts one more for the custom DNS server, which does not take a reserved address.

2. Answer C Concept 1 · transfer

You need an Azure subnet for 50 VMs. What is the smallest prefix that works?

Target word: solve for prefix **Decides:** /26 gives 59; /27 gives 27.

A	Inverted operation	Larger prefix means fewer addresses: 3 usable.
B	Adjacent category	Fits, but not the smallest.
C	Correct	$64 - 5 = 59 \geq 50$.
D	Dropped element	Only 27 usable.

3. Answer A Concept 2

An NSG has priority 200: deny all inbound from 203.0.113.0/24, and priority 300: allow inbound TCP 443 from any source. ~~The NSG was created last week and is attached to a subnet with four VMs.~~ A client at 203.0.113.50 opens HTTPS to a VM. What happens?

Target word: result of priority evaluation **Decides:** 200 is processed before 300.

A	Correct	Lower number is processed first and matches.
B	Adjacent category	Defaults apply only if nothing earlier matches.
C	Inverted operation	Treats higher numbers as higher priority.
D	Absolute language	Priority order decides.

4. Answer D Concept 2 · transfer

An NSG denies SSH from the internet at priority 200. You must allow SSH from one admin IP range. What priority should the new allow rule have?

Target word: solve for priority **Decides:** The allow must be processed before 200.

A	Wrong stem number	Far after the deny; and that range is reserved for defaults.
B	Confusion set	Priorities must be unique within a direction.
C	Inverted operation	The deny would match first.
D	Correct	It must be evaluated before the deny.

5. Answer B Concept 3

A help-desk analyst must restart and resize VMs in the rg-web resource group only. A teammate proposes Owner at the subscription so the analyst “never gets blocked.” ~~The analyst works the night shift.~~ What assignment fits?

Target word: role and scope **Decides:** Task is limited to VMs in one resource group.

A	Dropped element	Cannot restart or resize.
B	Correct	Narrowest role at the lowest scope.
C	Inverted operation	Broadest role at a broad scope.
D	Adjacent category	Scope reaches far beyond one group.

6. Answer B Concept 3 · transfer

A user has Contributor on resource group rg-app only. Can the user modify a storage account in rg-data in the same subscription?

Target word: reverse: scope limits **Decides:** Assignments inherit downward only.

A	Adjacent category	Region does not affect RBAC scope.
B	Correct	Scope does not spread sideways.
C	Inverted operation	Scope is where it was assigned.
D	Confusion set	They do not.

7. Answer C Concept 4

A web app in Azure App Service reads from Azure Key Vault. The current code stores a client secret in a config file. ~~The app is written in C# and deployed from a pipeline.~~ What is the better approach?

Target word: authentication method **Decides:** Azure-to-Azure access without stored secrets.

A	Camouflage	Language does not decide.
B	Dropped element	Still a stored secret.
C	Correct	No secret to store or rotate.
D	Inverted operation	Makes exposure worse.

8. Answer A Concept 5

A resource group holds a database, two VMs, and a storage account. Your Bicep template defines only the two VMs. You deploy it in complete mode. ~~The template file is 90 lines long.~~ What happens?

Target word: effect of complete mode **Decides:** Resources not in the template are deleted.

A	Correct	Complete mode removes resources not in the template.
B	Absolute language	The VMs are in the template and remain.
C	Confusion set	That is incremental mode.
D	Camouflage	Length does not decide.

9. Answer B Concept 6

Company rules say all resources must be in approved US regions and carry a cost-center tag. There are 30 people with Contributor rights, ~~and deployments run through two pipelines~~. What is the right tool to enforce the rules?

Target word: enforce resource properties **Decides:** The rule is about what, not who.

A	Inverted operation	Blocks work instead of setting rules.
B	Correct	Policy governs resource properties for everyone.
C	Confusion set	RBAC controls actions, not allowed regions or tags.
D	Camouflage	Pipelines do not enforce resource rules.

10. Answer D Concept 7

VNet-A is peered with VNet-Hub, and VNet-Hub is peered with VNet-C. No gateway or network appliance routes traffic in the hub. ~~The hub VNet is the oldest of the three~~. A VM in VNet-A cannot reach a VM in VNet-C. Why?

Target word: peering behavior **Decides:** Peering is not transitive.

A	Absolute language	Direct peers communicate.
B	Adjacent category	NSGs are not the cause here.
C	Camouflage	Age does not affect routing.
D	Correct	Traffic does not pass through a peered VNet by default.

11. Answer A Concept 7 · transfer

VNet-A and VNet-C must communicate, and the hub has no routing appliance. Which change makes it work?

Target word: reverse: fix **Decides:** Direct peering.

A	Correct	Direct peers can communicate.
B	Inverted operation	Removes connectivity.
C	Confusion set	Region does not decide peering reachability.
D	Dropped element	Does not connect A to C.

12. Answer D Concept 8

Someone deleted a load balancer yesterday. ~~The team's Azure Monitor dashboard shows CPU and network graphs, and Defender for Cloud gives the subscription a secure score of 72%~~. Where do you find who deleted it?

Target word: who performed a control-plane action **Decides:** Management operations with caller identity.

A	Confusion set	Records inside one VM, not Azure resource operations.
B	Camouflage	Performance data does not record callers.
C	Adjacent category	Posture, not an operation history.
D	Correct	Records control-plane operations and who performed them.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	5	
Wrong stem number	A figure copied from the prompt that is not the answer.	1	
Adjacent category	The right idea in the wrong bucket.	7	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	8	
Absolute language	Always, never, guarantee — where judgment applies.	3	
Confusion set	Every option is real; the test is exact role-matching.	6	
Camouflage	Answers a stem fact that was never the question.	6	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Azure subnet addressing	Azure reserves 5 addresses in every subnet, so usable addresses = $2^{(32 - \text{prefix})} - 5$.	Using the traditional “minus 2” rule in a VNet.	A VNet subnet prefix and a VM count.
2	NSG rule priority	NSG rules are processed from the lowest priority number up; the first matching rule decides, and default rules (65000+) apply only if nothing earlier matches.	Assuming a higher number means higher importance.	Numbered allow and deny rules that overlap.
3	RBAC scope and least privilege	Role assignments inherit downward (management group → subscription → resource group → resource); assign the narrowest built-in role at the lowest scope that does the job.	Granting Owner at the subscription for a single-resource-group task.	A task limited to one resource group or resource.
4	Managed identities	Azure resources should authenticate to other Azure services with a managed identity instead of secrets or connection strings stored in code.	Embedding a storage key or client secret in app settings or source code.	An Azure-hosted app calling another Azure service.
5	ARM/Bicep deployment modes	Incremental mode (default) adds or updates resources in the template and leaves others alone; complete mode deletes resources in the resource group that are not in the template.	Running complete mode against a shared resource group.	A resource group containing resources not in the template.
6	Azure Policy vs. RBAC	RBAC controls who can perform actions; Azure Policy controls what resource properties are allowed (for example, regions, SKUs, required tags), regardless of who deploys.	Trying to restrict allowed regions with a role assignment.	A rule about resource properties rather than people.
7	VNet peering is non-transitive	Peering connects two VNets directly; if A peers with B and B with C, A cannot reach C unless A and C peer or traffic is routed through a hub gateway or network virtual appliance.	Assuming peering chains pass traffic through.	Three VNets with two peerings.
8	Activity log vs. metrics vs. posture	The Activity log records control-plane operations and who performed them; Azure Monitor metrics track performance; Defender for Cloud assesses security posture.	Looking in performance metrics to find who changed a resource.	Who / how much / how secure.

Course scope (public description): In the BSCNE-Azure Capstone Project, students choose from a fixed set of tasks that they might encounter in the workplace. Examples might include configuring network components, applying security policies to a set of cloud resources, or automating a network deployment using orchestration tools. Students will submit their tasks for evaluation in a working, deployed environment where an evaluator can check their results by inspecting configurations or logging into machines or services to test functionality. All tasks will be completed in an Azure environment using either the Azure CLI or web-based tools and standard Azure platform components.